Dimple Yadav
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Projects

AI-Based Rockfall Early Warning System for Open-Pit Mines | *Python, Scikit-learn, TensorFlow* **Oct 25 - Dec 25**

- Developed predictive model for rockfall timing and location using multi-source data: geotechnical sensors (crack meters, strain gauges, earth pressure cells), micro seismic events, DEM/drone imagery, and environmental factors
- Engineered features from sensor time-series (displacement rates, strain trends, stress changes)
- Performed data fusion, cleaning, and EDA on heterogeneous datasets; built classification/regression pipeline for early warning alerts with high precision on simulated quarry data
- Designed real-time dashboard prototype for mine operators

Repository Link: **https://github.com/Dimple04152/RockWatch**

Nutrient Variability Analysis for Food Labeling | *Python, Pandas, Matplotlib, Seaborn* **Apr 25 – May 25**

- Analyzed USDA FoodData Central dataset (8,000+ items) to study macronutrient and calorie conversion variability across categories
- Identified outliers (e.g., Category 20 high calories/carbs) and method-dependent differences (Vitamin C sensitivity).
- Derived insights on labeling reliability and dietary implications

Repository Link: **https://github.com/Dimple04152/Nutrient-Analysis-FoodData-Central**

Maldives Tourism Performance Dashboard | *Power BI, Python (ETL)* **Nov 25 - Dec 25**

- Extracted and cleaned multi-year tourism data from PDF reports; built fact tables for arrivals, occupancy, and airlines
- Created interactive Power BI dashboard with slicers and visuals revealing 65.25% avg occupancy, 4M arrivals, and Emirates dominance (1.4M passengers)
- Enabled exploratory analysis of operational trends and facility distribution

Repository Link: **https://github.com/Dimple04152/tourism-etl-project**

Educational Insights Dashboard | *Excel, PivotTables, Slicers* **Apr 25 – May 25**

- Analyzed real-world Pune school enrollment dataset (6,075 schools); cleaned and modeled data for multi-dimensional insights
- Built interactive dashboard showing zonal distribution, management diversity, and operational stability (~90%)
- Uncovered key insights: Yerawada zone leads with 632 schools, ~90% operational stability, and MNC management dominates enrollment — supporting educational planning and equity analysis

Repository Link: **dimple-yadav-473278296_excel-datavisualization-education**

INTERNSHIP

INFOSYS Springboard | Python, FastAPI, PostgreSQL, Redis, SQL Alchemy, REST APIs **Jan 26 – Ongoing**

- Designed an **AI-powered stock screener backend** supporting natural language financial queries via **LLM → DSL → SQL pipeline**
- Implemented a **custom SQL compiler and execution layer** generating safe parameterized SQL from structured DSL queries
- Built **REST APIs for companies, portfolios, alerts, and AI queries** with JWT authentication and protected endpoints. Developed **data ingestion pipeline** integrating Alpha Vantage and Yahoo Finance APIs with parsing, normalization, and Redis caching

Lovely Professional University | C++, Trie Data Structure, Flask (REST API), JSON, Git & GitHub **Jun 25 – Jul 25**

- Designed and implemented a Trie data structure supporting insert, delete, prefix search with frequency-based ranking
- Loaded 50k+ English words from public JSON dataset; added case-insensitive search and auto-insertion of new prefixes
- Converted into a lightweight local HTTP server (zero dependencies) to serve suggestions via REST API

Certificates / Certifications

Agile with Atlassian Jira **Jan 26**

Coursera — Certificate Link

AI Fusion Competition- 1st position **Feb 25**

IIT Ropar — Certificate Link

Data Structures & Algorithms using C++ **Jun 25**

CPE LPU — Certificate Link

Computer Communications Specialization **Dec 24**

University of Colorado System via Coursera — Certificate Link

Skills

Languages: Python, C++, Java

Technologies/Frameworks: FastAPI, Redis, SQLAlchemy, REST APIs, Pandas, NumPy, Matplotlib, Seaborn, Power BI, Excel (PivotTables/Slicers), Flask, Git, GitHub, Agile Methodologies

Domain Skills: Machine Learning, Computer Vision, Exploratory Data Analysis, Statistics, Data Structures & Algorithms, Data Visualization

Databases: MySQL, MongoDB, PostgreSQL

Soft Skills: Analytical Problem-Solving, Collaborative Teamwork, Proactive Project Management, Agile Adaptability

Education

Lovely Professional University **2023 – 2027**

Bachelor of Technology - Computer Science and Engineering — CGPA: 7.93 *Phagwara, Punjab*

Kendriya Vidyalaya No. 1, Jalandhar Cantt **Apr 20 – Mar 22**

12th — Percentage: 88.0% *Punjab*

Kendriya Vidyalaya No. 1, Jalandhar Cantt **Apr 19 – Mar 20**

10th — Percentage: 93.0% *Punjab*